

Line-Following Recovery Logic

Program a rover to follow a path and recover when the line is temporarily lost.

Advanced**2 class periods****Engineering Challenge**

Engineering Context

A warehouse-style rover transports equipment through a marked route. Dust or damaged floor markings may interrupt the path, so the rover needs recovery behavior.

What You Will Practice

- Use feedback control
- Create state-based logic
- Test fault recovery
- Measure path performance

What You Need

- Rover with optical or line sensors
- Marked course with one intentional gap
- Programming environment
- Stopwatch
- Engineering notebook

Student Instructions

- 1 Create baseline line-following logic and test it on a continuous path.
- 2 Define the conditions that indicate the line has been lost.
- 3 Design a recovery sequence such as stop, search left, search right, and timeout.
- 4 Add a safe stop if the line cannot be found.
- 5 Run five trials through a path containing a gap.
- 6 Record recovery success, completion time, and any departures from the lane.
- 7 Tune the logic and repeat five official trials.

Engineering Requirements

- Follow the normal path without manual control
- Recover from the designated gap
- Stop safely after the search timeout
- Complete at least four of five official trials
- Code must include meaningful comments

Constraints & Safety

- Use low operating speed during development
- Maintain a clear test area
- Provide immediate stop control
- Do not allow uncontrolled searching outside the boundary

What You Submit

- State diagram
- Commented code
- Baseline and final results
- Recovery success rate
- Tuning explanation

Reflection Questions

- 1 What event caused the rover to enter recovery mode?
- 2 Which search pattern worked best?
- 3 How did timeout length affect safety?
- 4 What additional sensor could improve recovery?

Engineering Tip Draw the control states before writing code: Follow, Lost, Search Left, Search Right, Stop.	Common Mistake Adding more nested conditionals without a state plan can make recovery behavior unpredictable.
Stretch Goal Add a second fault such as an obstacle that interrupts the path.	Career Connection Autonomy engineers design fault-handling states so robotic vehicles remain safe when sensors or navigation references are lost.